Lecture VIII - Police Districting

Applied Optimization with Julia

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Introduction

Police Service District Planning

T. Vlćek, K. Haase, M. Fliedner, and T. Cors [1]

Learning Objectives

After this lecture, you can:

- Explain the territory design problem and its applications
- Model districting as a **p-median problem**
- Explain the difference between **contiguity** and **compactness**
- Enforce both with additional constraints
- Discuss the assumptions and limitations of the model

Challenges

Question: **What makes the work of emergency services complex?**

- Dynamic urban development
- Changing population patterns
- Resource constraints
- Need for rapid response
- Multiple stakeholder interests

Emergency Services

Emergency services address the needs of three interest groups:

- Citizens
- Service personnel
- Administrators

Question: **What could be the objectives of these groups?**

Stakeholder Objectives

1. **Citizens**
   - Fast response times
   - Reliable service coverage
2. **Service Personnel**
   - Manageable workloads
   - Safe working conditions
3. **Administrators**
   - Cost efficiency
   - Resource optimization

> ℹ Note
>
> Aligning the objectives of the three interest groups is challenging.

Emergency Service Districting

Question: **Why might current district layouts be suboptimal?**

- Many layouts date back several decades
- Often designed along highways and regions [2]
- Extensive data **not used** for data-driven improvement

How can we improve

this situation?

The Role of Data

Question: **What data can help improve emergency services?**

- Historical incident patterns
- Response time analysis
- Resource utilization metrics
- Population densities and traffic patterns

. . .

> ℹ Note
>
> Extensive data collected, but often lack of tools or knowledge to leverage it.

Optimization

- Operations research (OR) models can help!
- Based on **incident records** and **geographical information**
- Improve the **response of emergency services**
- Help administrators in making **strategic decisions**
- Locate new departments or close departments [3]

Case Studies

Police Districting

> For an efficient and effective distribution of resources, police jurisdictions are divided into precincts or command districts with separate departments. These are further divided into patrol beats [4].

Service Priority Extremes

- **High Priority**
  - Life-threatening situations
  - Active crimes in progress
  - Multiple unit response needed
- **Low Priority**
  - Minor incidents
  - Administrative tasks

Case Studies

- Different urban contexts
- Study of jurisdictions in
  - Germany: Large metropolitan area
  - Belgium: Large rural area
- Focus on response time optimization

. . .

> **Note**
>
> Part of the force patrols the streets, another part is **stationed at the departments**.

Dispatching

- Dispatchers assign all calls for service (CFS) to vehicles from the corresponding districts and patrol areas
- Officers are **familiar with the area** and are thus better prepared to respond appropriately [5]
- To cope with high demands, dispatchers can **assign vehicles from nearby districts or beats**

Potential Problem

Question: **What could be the potential problem?**

. . .

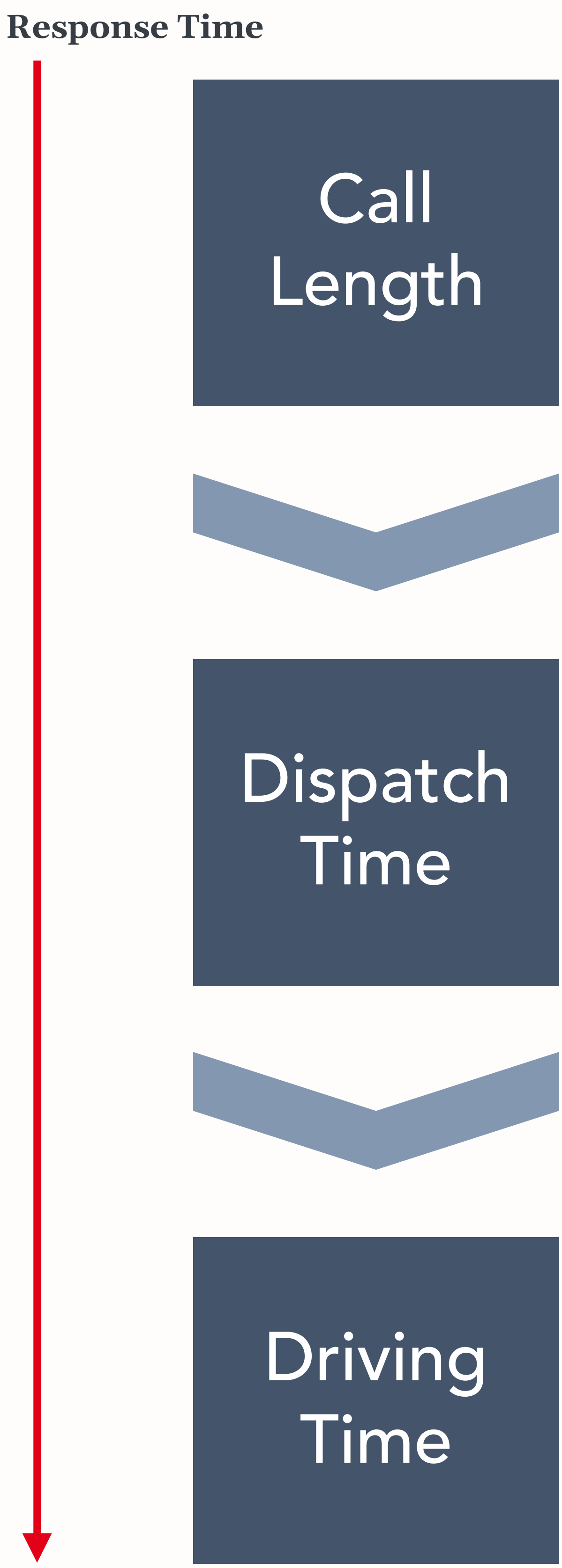
- This can lead to a domino effect
- Transferring vehicles from other districts or beats **reduces coverage in those locations** [6]
- This makes them vulnerable to missing resources when **they need assistance themselves**

Overloaded Systems

- This can lead to overloaded systems!
- Long dispatching delays due to **staff shortages**
- Preventive patrol hardly possible [7]
- Dispatchers **constantly** draw on patrol resources
- Reduces the **response time** of emergency services

. . .

> **Warning**
>
> This is a **common problem** in many emergency services.


Response Time
Call Length
Dispatch Time
Driving Time

- Central criterion to measure the effectiveness of emergency services is the response time
- Time between a **call for aid** and the arrival at the **incident location**
- Low response time increases the likelihood of helping and improves confidence [5]

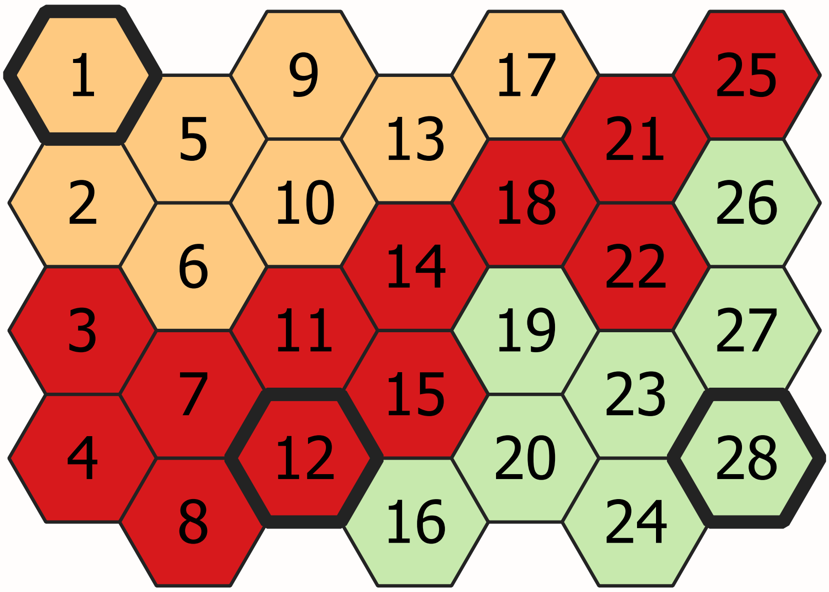
Response Time Influencers

Question: **What affects response time?**

- Initial contact
- Information gathering
- Unit assignment
- Resource coordination
- Route to location
- Traffic conditions

Territory Design Problem

What is Territory Design?



Aggregation of **small geographic areas**, called basic areas (BAs), into geographic clusters, called districts, so that these are acceptable according to **pre-defined planning criteria**[1].

. . .

[1]A. A. Zoltners and P. Sinha [8]

ℹ Note

Territory design is a general framework applicable to many domains: emergency services, sales territories, political districts, school districts, etc.

Territory Design Components

Question: **What are the key components?**

- **Basic Areas (BAs)**: Smallest indivisible geographic units
- **Districts/Territories**: Aggregation of BAs
- **Planning Criteria**: Rules and objectives for grouping
- **Decision**: Which BAs belong to which district?

. . .

Tip

In our police case: BAs are hexagonal cells, districts are service areas for police departments.

Common TD Objectives

Balance Criteria:

- Equal workload distribution
- Equal sales potential
- Equal population coverage
- Resource balance

Geographic Criteria:

- Contiguity: No isolated basic areas
- Compactness: "Round", undistorted shapes
- Minimize travel distances
- Respect boundaries

. . .

! Important

These criteria often conflict with each other… Optimization helps find the best trade-off!

Applications of TD

Question: **Where else is this used?**

- **Sales force deployment**: Sales reps to territories
- **Political districting**: Electoral boundaries

- **School districts**: Student assignment zones
- **Waste collection**: Service route planning
- **Healthcare**: Hospital catchment areas

. . .

> **Note**
>
> Same mathematical structure, different objectives and constraints!

Police Districting Specifics

Question: **What could be the objective?**

. . .

- Minimize the response time to help citizens faster while **increasing the confidence in the service**

. . .

Question: **What could be further objectives?**

. . .

- Reallocate **only part** of the police department's resources
- Compact and contiguous territories to **improve patrol**
- Prevention of **isolated departments**

Response Time Components


Length
Dispatch Time
Driving

Question: **How can we model response time?**

- Call length is independent of territory
- Dispatch time is **difficult to model**
- Driving time can be **minimized directly**

⚠ Conclusion

We focus on **minimizing expected driving times** between departments and incident locations.

Let's build our model

step by step!

Model Formulation

Key Model Components

Question: **What could be our key model components?**

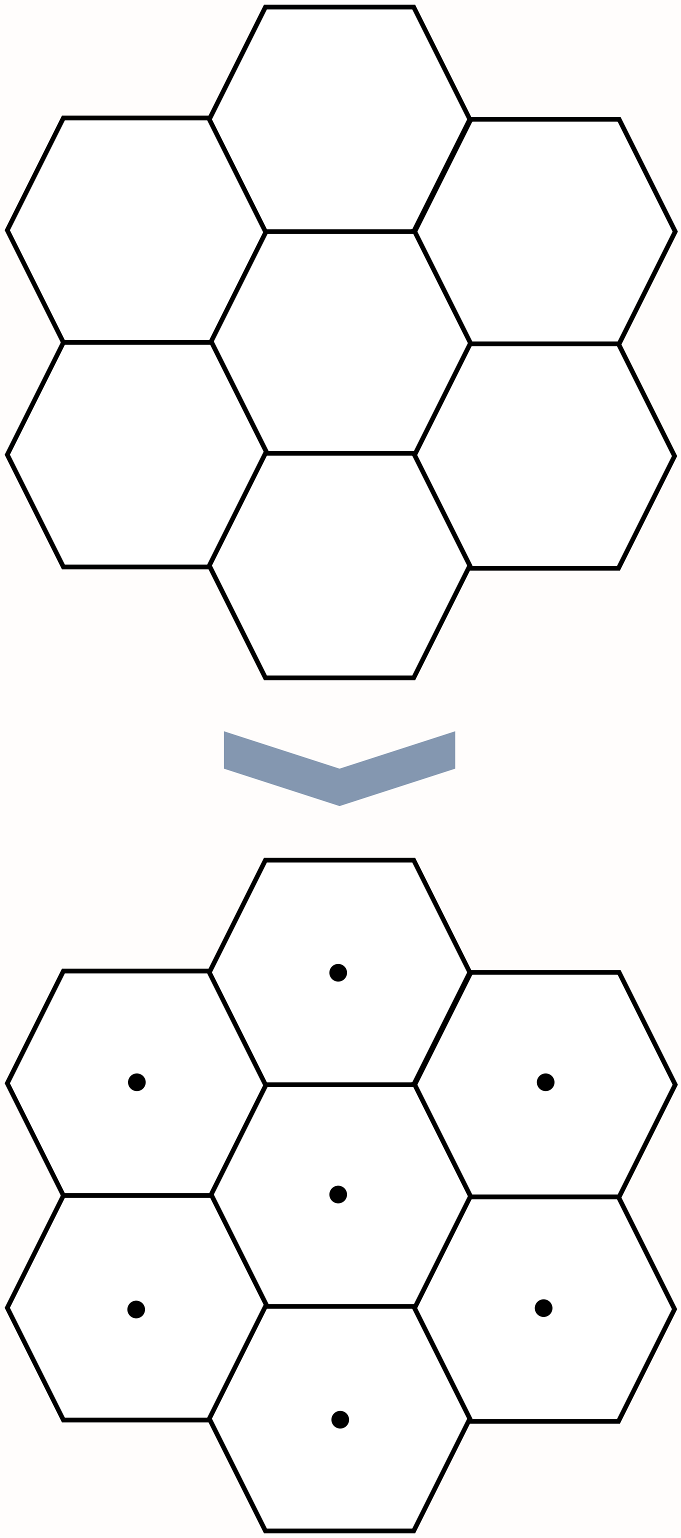
...

- Basic areas (BAs) and potential department locations
- Driving times between department locations and BAs
- Forecasted incident data
- Assignment decisions

...

Question: **Which are sets, parameters, and variables?**

From Geography to Graph



Question: **How do we model this mathematically?**

- Geographic areas → Centroids
- Centroids → **Vertices** in a graph

- Spatial relationships → **Edges** between **vertices**

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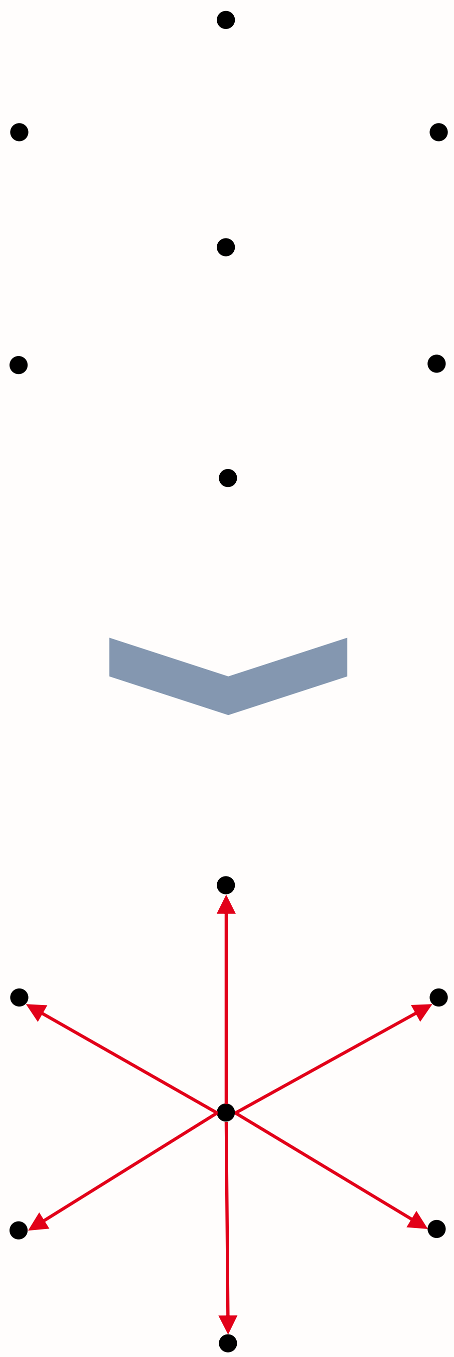
> 💡 Tip
>
> This abstraction allows us to use optimization techniques from graph theory and location science!

Mathematical Structure



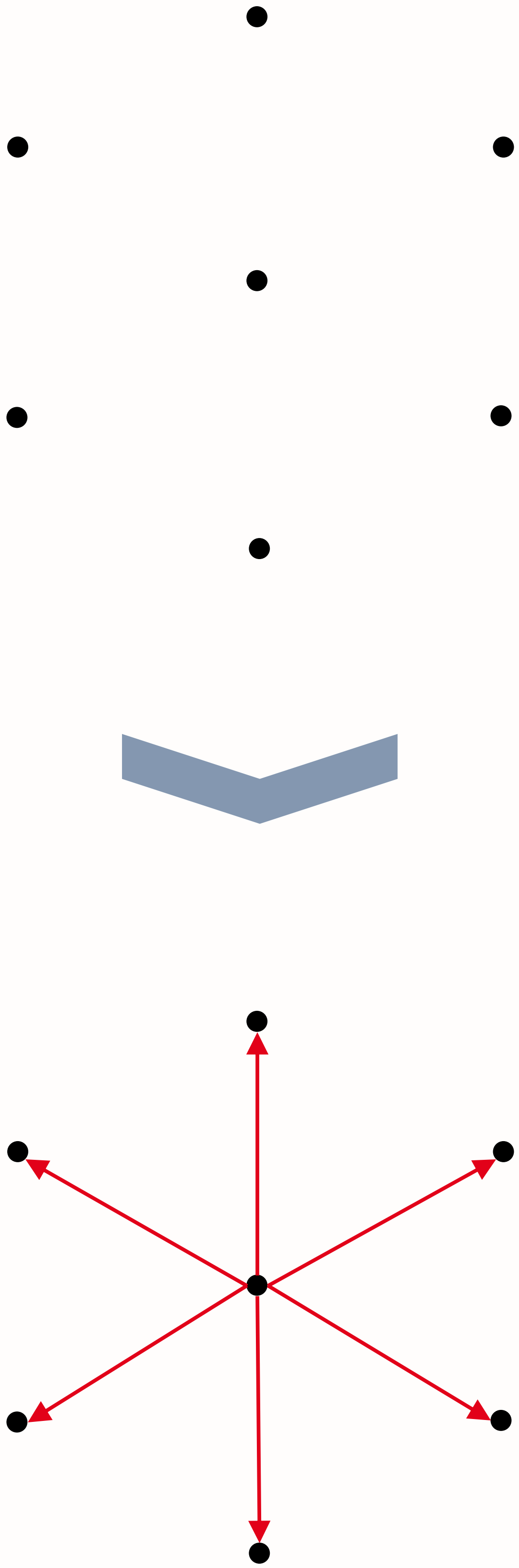
Question: **What sets do we need?**

- $\mathcal{J}$: Set of **all BAs** (basic areas), indexed by j
- $\mathcal{I}$: Set of **potential department locations** where $\mathcal{I} \subseteq \mathcal{J}$, indexed by i

...

> **! Important**
>
> Department locations are a subset of all BAs as not every BA can host a department!

Why Hexagons?



Question: **Advantages of hexagons?**

- Equal distances to all neighboring centroids
- Reduces sampling bias from edge effects [9]
- Special properties that help with the enforcement of **compactness**
- Better representation of urban geography

Parameters

Question: **What parameters do we need?**

- p - Number of departments to be located
- $t_{i,j}$ - Expected driving time from department location i to BA j
- w_j - Forecasted number of incidents in BA j

. . .

> Tip
>
> Parameters should be carefully calibrated with real-world data!

Decision Variable(s)?

> i We have the following sets:
>
> - BAs, indexed by $j \in \mathcal{J}$
> - Potential department locations, indexed by $i \in \mathcal{I}$

. . .

> ! Our objective is to:
>
> Minimize the expected response time of the emergency services by optimizing the assignment of BAs to departments.

. . .

Question: **What decisions do we need to model?**

Decision variable/s

- $X_{i,j}$: 1 if BA j assigned to department i, 0 otherwise

. . .

Question: **What is the domain of our decision variable?**

. . .

- $X_{i,j} \in \{0,1\} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}$

Let's build our

objective function!

Objective Function?

> ! Our objective is to:
>
> Minimize the expected response time of the emergency services by optimizing the assignment of BAs to departments.

...

Question: **How do we minimize response time?**

- We want to minimize **total driving time**
- Weight each BA by its forecasted incidents w_j
- No fixed opening costs: the **number** of departments is fixed to p anyway

Objective Function

Question: **What could be our objective function?**

...

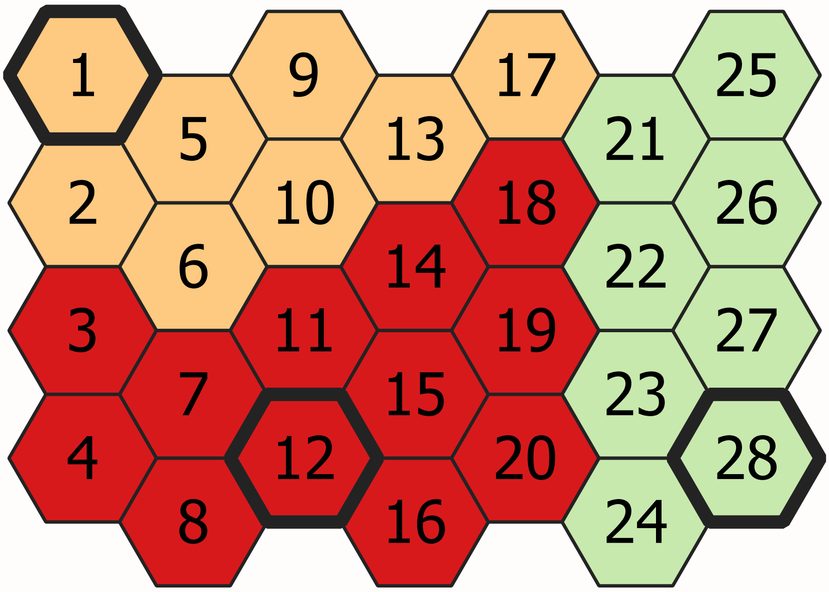
$$\text{Minimize} \quad \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} w_j \times t_{i,j} \times X_{i,j}$$

...

> ! Expected Driving Time
>
> - Total driving time **across all assignments**
> - Weighted by the **incident frequency** w_j of each BA
> - Considers **all possible BA-department pairs**

Constraints

Key Constraints



Question: **Constraints needed?**

1. BA must have one department
2. Limit **number of departments**
3. Only assign **active** departments
4. Ensure **contiguous** districts
5. Maintain district **compactness**

Single Assignment Constraint?

Question: **Why do we need this constraint?**

. . .

- Each BA must be assigned to exactly one department
- Prevents **overlapping** jurisdictions
- Ensures **complete coverage**

. . .

ℹ We need the following variables:

- $X_{i,j}$ - 1 if BA j assigned to department i, 0 otherwise

Single Assignment Constraint?

Question: **What could the constraint look like?**

. . .

$$\sum_{i \in \mathcal{I}} X_{i,j} = 1 \quad \forall j \in \mathcal{J}$$

. . .

i Note

Each BA must be assigned to exactly one department.

Department Count Constraint?

! The goal of these constraints is to:

Ensure that exactly p departments are opened.

. . .

i We need the following sets, parameters and variables:

- $\mathcal{I}$ - Set of potential department locations, indexed by i
- $X_{i,j}$ - 1, if BA j assigned to department i, 0 otherwise
- p - Number of departments

The Self-Assignment Trick

Question: **What does the variable $X_{i,i}$ stand for?**

. . .

- BA i is assigned to the department in **the same BA** i
- An open department **always serves its own BA**
- Hence: $X_{i,i} = 1$ exactly if department i is open

. . .

i Note

Many textbooks use a separate variable Y_i to model opening a facility. Using the "diagonal" $X_{i,i}$ instead saves these extra variables — both formulations are equivalent here.

Department Count Constraint

Question: **What could the constraint look like?**

. . .

$$\sum_{i \in \mathcal{I}} X_{i,i} = p$$

. . .

Question: **What happens if we have more departments than potential locations?**

. . .

- We **can't open more departments** than there are locations
- The model **will be infeasible**

Active Department Constraint?

! The goal of these constraints is to:

Ensure that each BA is assigned to an active department, e.g. a department that is opened and that could dispatch vehicles.

. . .

i We need the following sets and variables:

- $X_{i,j}$ - 1, if BA j assigned to department i, 0 otherwise

Question: **How do we ensure assignments only to active departments?**

Active Department Constraint

$$X_{i,j} \leq X_{i,i} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}$$

. . .

i Note

This constraint creates a **logical connection between department locations and BA assignments** where BAs can only be assigned to opened departments.

p-Median Problem

Connection to Facility Location

Question: **How does this relate to these problems?**

. . .

p-Median Problem:

- Fixed number of facilities (exactly p)
- Minimize total distance
- All customers served
- No capacity constraints

Similar problems:

- **UFLP**: Uncapacitated Facility Location
- **p-Center**: Minimize maximum distance
- **Capacitated**: Add limits

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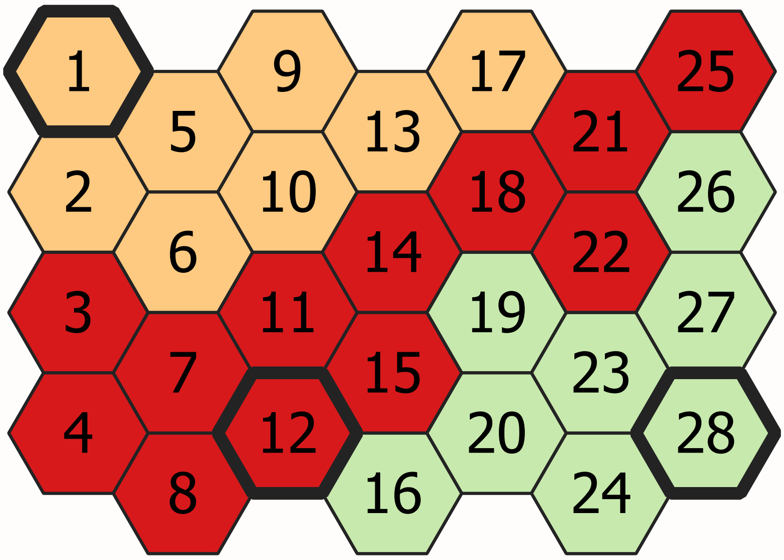
ℹ Note

Our police districting is a **p-Median variant** with additional geographic constraints (contiguity, compactness)!

Contiguity and Compactness

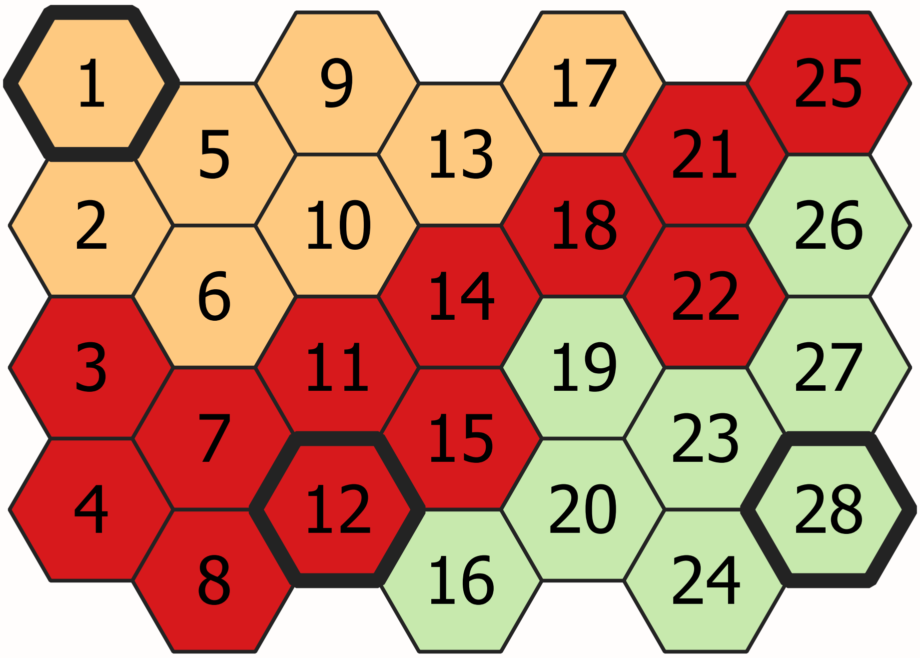
Contiguity Introduction

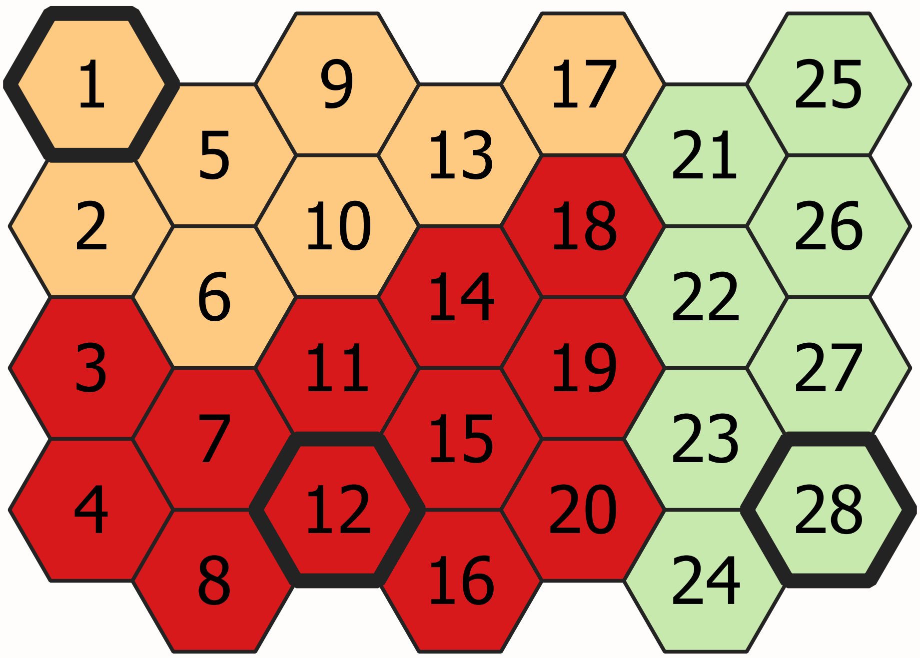


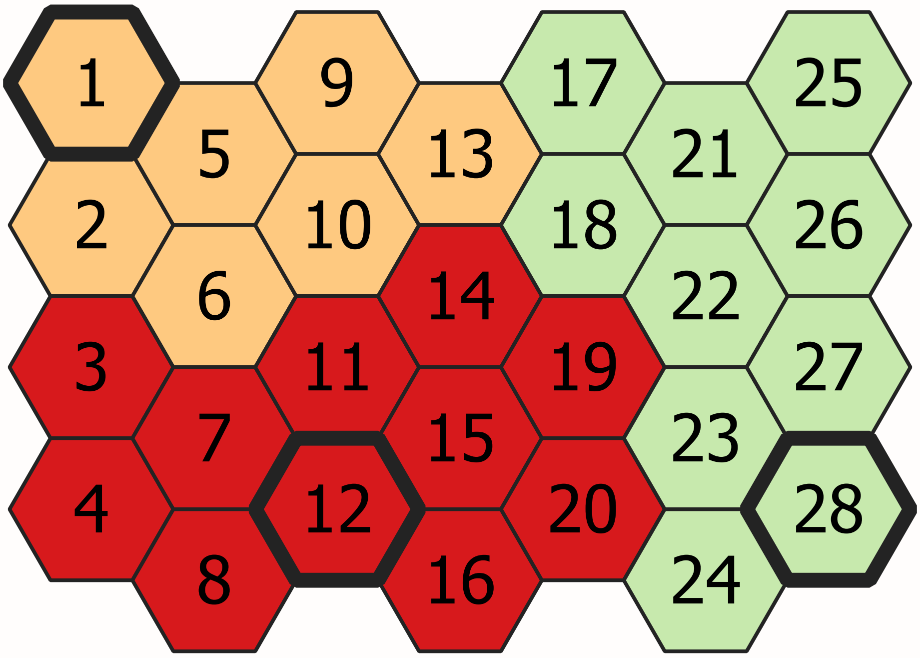
Question: **Why is contiguity important?**

- Prevents **isolated areas**
- Ensures **contiguous patrol routes**
- Maintains **operational coherence**

What is compactness?





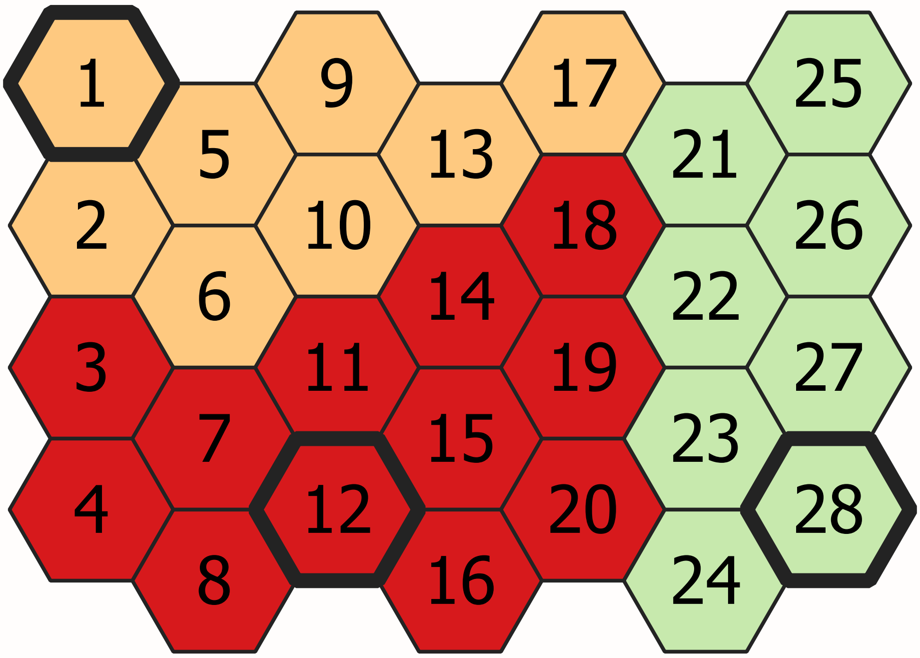


. . .

i Compactness

Compactness has **no univocal definition**; a district is commonly declared compact if it is 'somehow round-shaped and undistorted' [10].

Contiguity and Compactness



Question: **Are our resulting districts based on the model contiguous and compact?**

- This depends on $t_{i,j}$
- **If Euclidean distance**
  - Districts will be contiguous
  - Likely of compact shape

Compactness p-Median

Question: Is this likely for police service districting?

- No, as we minimize the driving time within a city
- Highways, Tunnels, etc.
- Multiplied by the differing number of requested cars
- This can contribute to distorted district shapes

Contiguity Sets

Additional Set and Parameter

- $e_{i,j}$ - Euclidean distance between centroids
- $\mathcal{A}_j$ - Set of BAs adjacent to BA j

...

$$\mathcal{N}_{i,j} = \left\{v \in \mathcal{A}_j \mid e_{i,v} < e_{i,j}\right\} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}$$

. . .

> **i The idea**
>
> BAs **closer to department** i than BA j in Euclidean distance and **adjacent to** j!

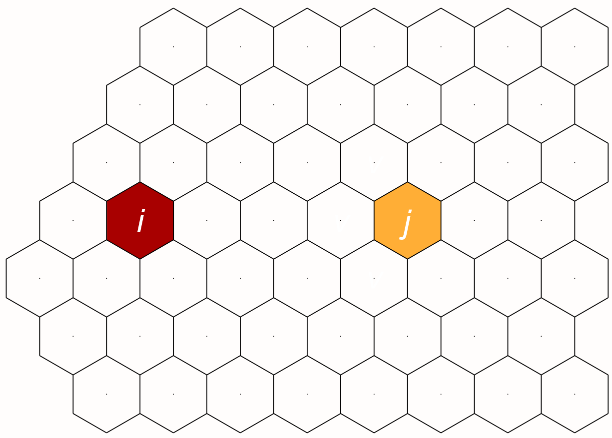
Careful with Ties!

- If all adjacent BAs are *exactly* as far from i as j, the strict inequality leaves $\mathcal{N}_{i,j}$ empty
- Then the assignment of j to i is **forbidden**
- Implementations should break ties, e.g. with $e_{i,v} \leq e_{i,j}$ and $v \neq j$

. . .

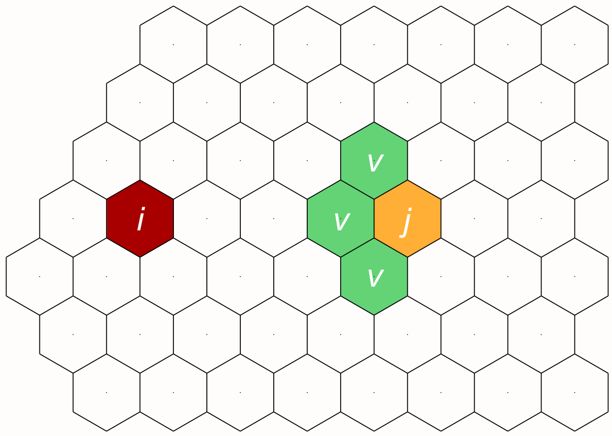
> **i Note**
>
> On a complete hexagonal grid, this cannot happen: one neighbor of j is always strictly closer to i. But real-world maps with holes, concave boundaries, or irregular BAs can produce ties!

Example A



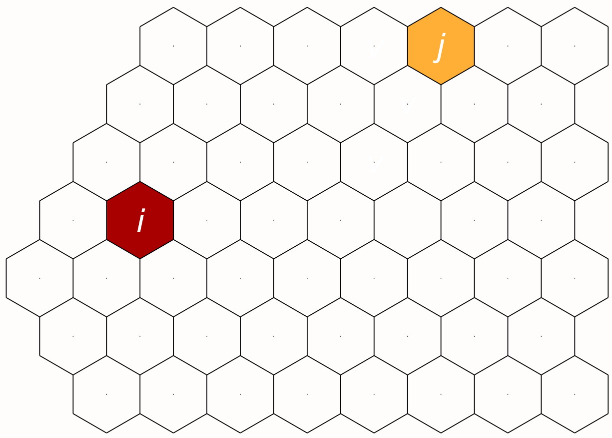
Department i and BA j: which neighbors of j are **closer to** i?

Example A



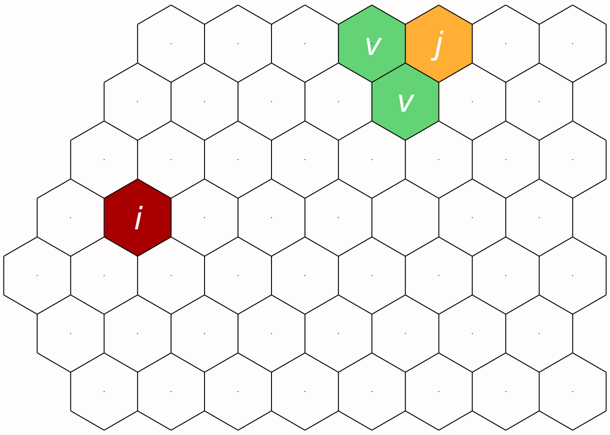
All three highlighted BAs v are adjacent to j **and** closer to i: $|\ \mathcal{N}_{i,j}\ | = 3$.

Example B



Same department i, but BA j now lies at the **edge** of the grid.

Example B



Here, only two neighbors of j are closer to i: $|\ \mathcal{N}_{i,j}\ | = 2$.

Enforcing Contiguity

All districts have to be contiguous

$$X_{i,j} \leq \sum_{v \in \mathcal{N}_{i,j}} X_{i,v} \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \setminus \mathcal{A}_i : i \neq j$$

...

> ! The idea
>
> If BA j is assigned to department i, then at least **one BA** that is adjacent to j and closer to i must **also** be assigned to department i!

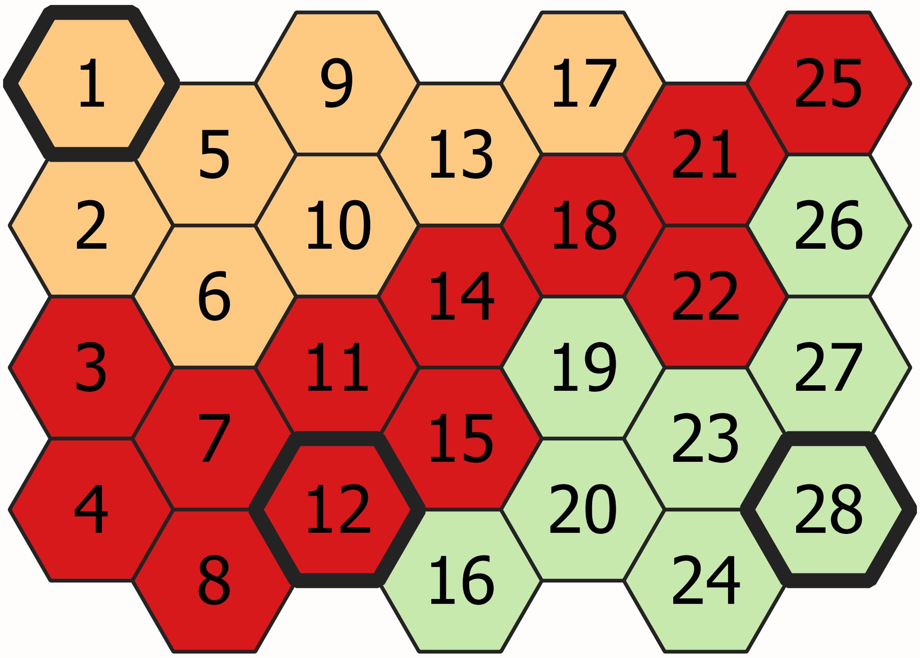
Contiguity and Compactness

All districts have to be contiguous and compact

...

> ! The idea
>
> If BA j is assigned to department i, then at least **two BAs** that are adjacent to j and closer to i must **also** be assigned to department i — if $|\ \mathcal{N}_{i,j}\ | > 1$!

Comparison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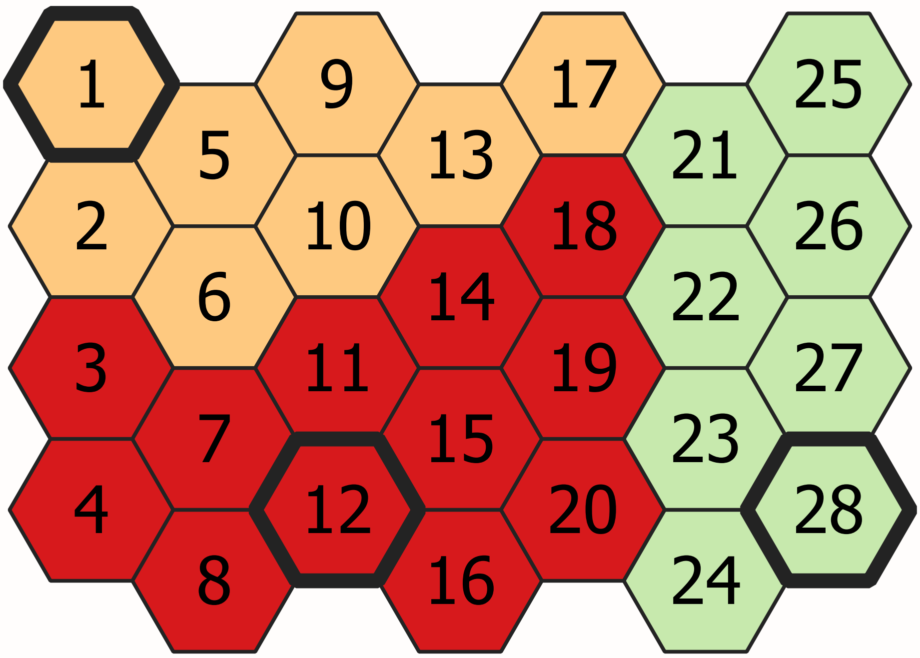
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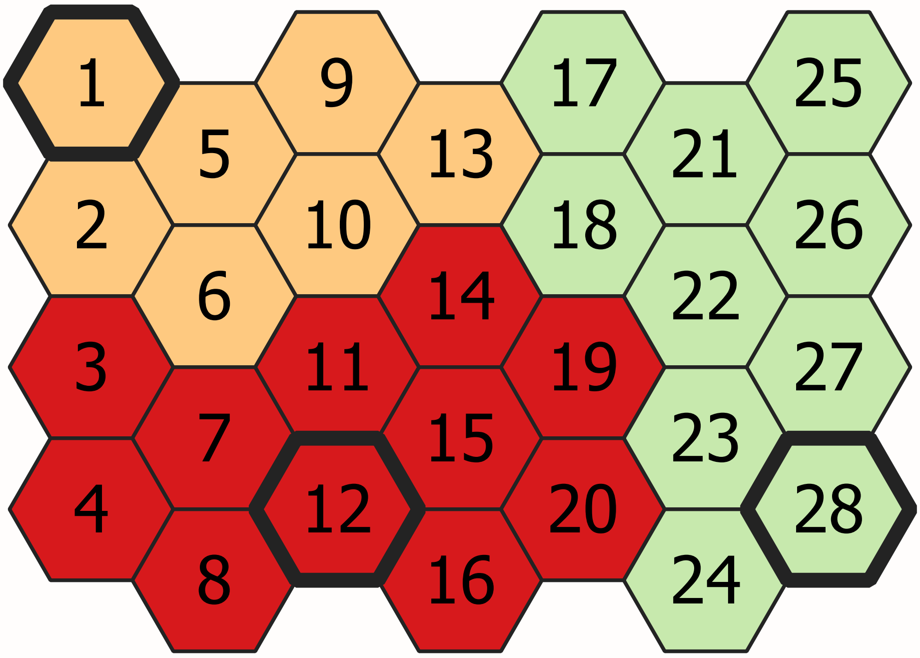

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> **i Why does this work?**
>
> Due to the constraints, there is **always a path back** to the department if a BA is assigned to a department! Contiguity alone still allows the “snake” on the left; the compactness constraint cuts off such shapes (middle, right).

Model Characteristics

Characteristics I

Question: **Is the model formulation linear or non-linear?**

. . .

- Objective and all constraints are **linear**
- With binary variables → mixed-integer problem (MIP)

. . .

Question: **What kind of variable domains do we have?**

. . .

- $X_{i,j}$ is **binary**: $|\mathcal{I}| \times |\mathcal{J}|$ variables in total

Characteristics II

Question: **Can the model be solved quickly?**

. . .

- For instances of **moderate size**: yes, with modern solvers
- The contiguity constraints add many rows → longer runtimes

. . .

Question: **Have we prevented isolated districts?**

. . .

- The plain p-median does **not** prevent them
- That is exactly why we added the contiguity constraints!

Model Assumptions

Questions: **On model assumptions**

- What assumptions have we made?
- Use Euclidean distances to approximate driving time?
- Can we rely on incident data collected by the police?

Implementation and Impact

Overview of Studies

Question: **Where did we apply our model?**

- Two distinct environments:
  1. Large metropolitan area (Germany)
  2. Rural region (Belgium)
- Different challenges and requirements
- Focus on response time optimization

German Metropolitan Case

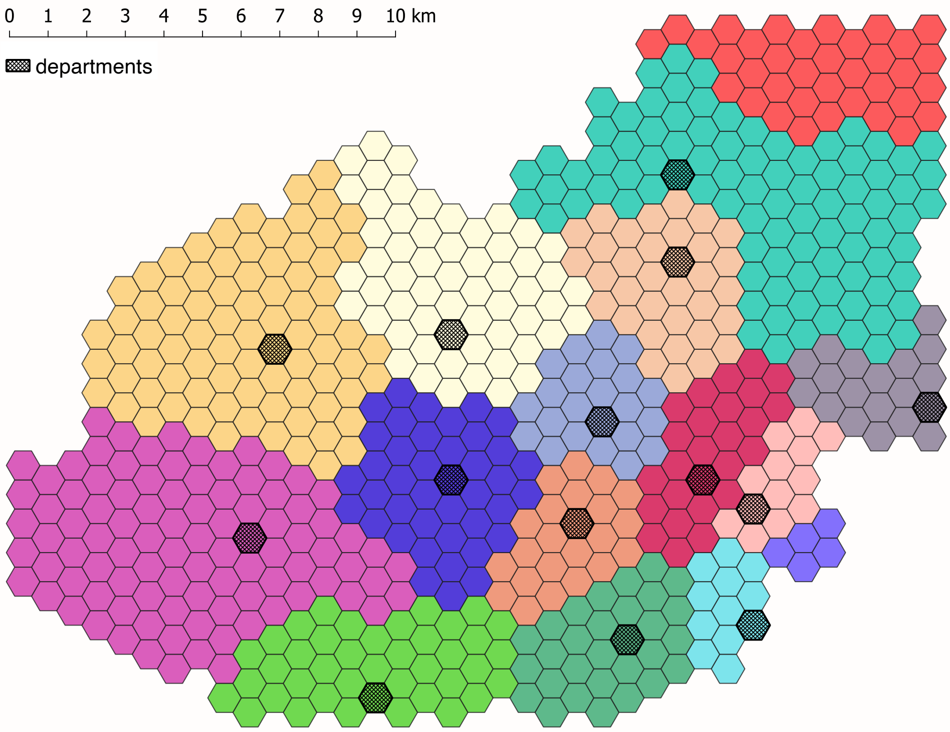
. . .

- 1.8 million incidents (2015-2019)
- ~20 department locations
- 1,596 basic areas
- Dense urban environment

. . .

Goal: Redesign districts to improve response time.

German Metropolitan Results



Belgian Rural Case

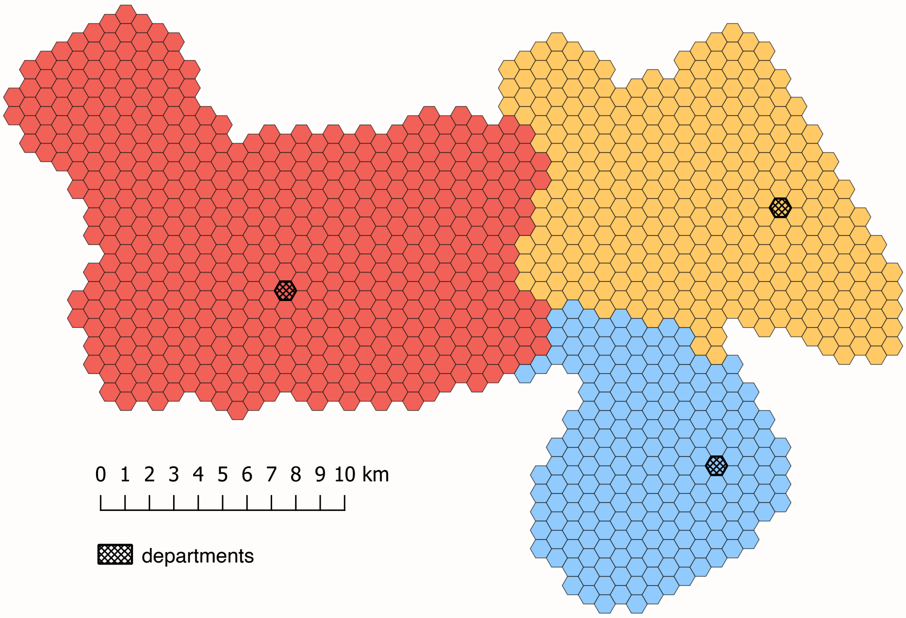
. . .

- 50,000 incidents (2019-2020)
- 2 existing + 1 planned location
- 1,233 basic areas
- Dispersed rural setting

. . .

Goal: Optimize coverage with limited resources.

Belgian Rural Results



Simulation Framework

How did we validate the results?

. . .

- Spatial and temporal patterns
- Shift schedules
- Priority handling
- Rush hours
- Inter-district support
- Variable driving times

Results

- Response time reduction up to 14.52%
- Better **workload distribution**
- Improved **coverage equity**
- More efficient **resource utilization**

. . .

> ! Important
>
> All improvements are **without additional staff**!

What About Workload Balance?

Question: **Our model only minimizes driving time. Where does the better workload distribution come from?**

...

- The model does **not** balance workload directly — the effect is emergent
- Compact districts around well-placed departments also **spread the expected workload** more evenly
- This was measured in the **simulation**, not enforced by a constraint

...

> Tip
>
> Think about it: how could you add an explicit workload balancing constraint with our parameters w_j and $t_{i,j}$?

Conclusions

1. Model **adaptability** crucial
2. **Local** context matters
3. Stakeholder buy-in essential
4. Data quality critical

...

> Tip
>
> Success requires balancing theoretical optimization with practical constraints!

Future Applications

Question: **Where else could this approach be useful?**

- Other emergency services
- Different urban contexts
- Resource allocation problems
- Service territory design

...

> Note
>
> The methodology is adaptable to various public service optimization scenarios.

Wrap Up

> And that's it for today's lecture!
>
> We now have covered districting problems and are ready to start solving some tasks in the upcoming tutorial.

Questions?

Literature

Literature I

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

Bibliography

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